

SIMATS ENGINEERING
CO1 - ASSESSMENT TOOL 1

COURSE NAME: Deep Learning

COURSE CODE: MLA0403

Name of the Student	V Sunil Kumar
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COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

CO1 Weightage: 15%

Assessment Tool Weightage: 35%

Duration: 1 Hour

Marks: 50

Code of Conduct:

*I V Sunil Kumar/192425292 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*



Signature of the student:

Name of the student : V Sunil Kumar

Duration: 1 Hour

Total Marks: 50

QUESTIONS:

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:

H H T H H H T H H T

Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).

Answer:

Given Outcomes:

H H T H H H T H H T

Step 1: Identify the total number of trials

Total coin tosses (**n**) = 10

Step 2: Count the number of Heads

Heads = 7

Step 3: Write the MLE formula

For a Bernoulli/Binomial experiment

$\hat{p} = \text{Number of Successes} / \text{Total Number of Trials}$

where:

- Success = Getting a Head
- Number of successes = 7
- Total trials = 10

Step 4: Substitute the values

$$\hat{p} = \frac{7}{10}$$

The Maximum Likelihood Estimate (MLE) of the probability of getting Heads is 0.7.

2. Out of 50 students, 42 passed an exam.

Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

Answer:

Given

- Total students = 50
- Students passed = 42

Step 1: Define the success

Success = Student passes the exam.

Step 2: Apply the MLE formula

$$\hat{p} = \frac{\text{Number of Successes}}{\text{Total Observations}}$$

Step 3: Substitute the values

$$\hat{p} = \frac{42}{50}$$

Step 4: Calculate

$$\hat{p} = 0.84$$

Answer

The MLE estimate of the probability of passing is 0.84 (84%).

3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective. Find the MLE for the probability that a bulb is defective.

Answer:**Given**

- Total bulbs inspected = 100
- Defective bulbs = 8

Step 1: Define the success

Success = A bulb is defective.

Step 2: Apply the MLE formula

$$\hat{p} = \frac{\text{Number of Defective Bulbs}}{\text{Total Bulbs}}$$

Step 3: Substitute values

$$\hat{p} = \frac{8}{100}$$

Step 4: Calculate

$$\hat{p} = 0.08$$

Answer

The Maximum Likelihood Estimate of the defective probability is 0.08 (8%).

4. A biased die is rolled 60 times. The number 6 appears 18 times. Estimate the probability of rolling a 6 using MLE.

Answer:**Given**

- Total rolls = 60
- Number of times 6 appears = 18

Step 1: Define success

Success = Rolling a 6.

Step 2: Apply the MLE formula

$$\hat{p} = \frac{\text{Number of Successes}}{\text{Total Trials}}$$

Step 3: Substitute values

$$\hat{p} = \frac{18}{60}$$

Step 4: Calculate

$$\hat{p} = 0.30$$

Answer

The MLE estimate of rolling a 6 is 0.30 (30%).

5. Given:

Input (x) = 4

Weight (w) = 2

Actual Output = 10

Prediction:

$$y = w \times x$$

1. Find the predicted output.
2. Calculate the error.
3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Answer:

Given

- Input (x) = 4
- Weight (w) = 2
- Actual Output = 10
- Prediction Formula:

$$y = w \times x$$

- Gradient = 4
- Learning Rate = 0.01

(a) Find the Predicted Output

Formula

$$y = w \times x$$

Substitute the values

$$\begin{aligned} y &= 2 \times 4 \\ y &= 8 \end{aligned}$$

Answer

Predicted Output = 8

(b) Calculate the Error

Formula

$$\text{Error} = \text{Actual Output} - \text{Predicted Output}$$

Substitute values

$$\begin{aligned} \text{Error} &= 10 - 8 \\ \text{Error} &= 2 \end{aligned}$$

Answer

Error = 2

(c) Update the Weight

In Gradient Descent, the weight is updated using:

$$w_{\text{new}} = w - \eta \times \text{Gradient}$$

where

- $w = 2$
- Learning Rate (η) = 0.01
- Gradient = 4

Step 1: Multiply the learning rate and gradient

$$0.01 \times 4 = 0.04$$

Step 2: Update the weight

$$w_{\text{new}} = 2 - 0.04$$

$$w_{\text{new}} = 1.96$$

Answer

Updated Weight = 1.96

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

